kibr $k_{\mathrm{ibr}}$ IBR current ratio, $I_{\mathrm{IBR}}/I_{\mathrm{rated}}$ Idiff $I_{\mathrm{diff}}$ Differential operating current Ibias $I_{\mathrm{bias}}$ Bias (restraint) current Iop $I_{\mathrm{op}}$ Relay operating (pickup) current km $k_m$ Adaptive bias slope of 87L characteristic Zline $Z_{\mathrm{line}}$ Line positive-sequence impedance Zapp $Z_{\mathrm{app}}$ Apparent impedance seen by distance relay alpha $\alpha$ Per-unit fault location along protected line (0 = local end) TMS $\boldsymbol{TMS}$ Time-Multiplier Setting for IEC inverse-time overcurrent relay CTI $\boldsymbol{CTI}$ Coordination Time Interval between primary and backup relay M $M$ Relay current multiple, $I_{\mathrm{fault}}/I_{\mathrm{pickup}}$ lambda $\lambda$ Forgetting factor in online SGD ($0 < \lambda \leq 1$) S $S_k$ CUSUM score at sample $k$ delta $\delta$ CUSUM reference value (allowable slack) theta $\theta$ CUSUM alarm threshold P $\mathbf{P}$ EKF error covariance matrix K $\mathbf{K}$ EKF Kalman gain matrix Q $\mathbf{Q}$ EKF process noise covariance R $\mathbf{R}$ EKF measurement noise covariance

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# THESIS CERTIFICATE

This is to undertake that the Thesis titled **TEST TITLE**, submitted by me to the Indian Institute of Technology Madras, for the award of **Doctor of Philosophy**, is a bona fide record of the research work done by me under the supervision of **Dr. Test**. The contents of this Thesis, in full or in parts, have not been submitted to any other Institute or University for the award of any degree or diploma.

**Chennai 600036** **Test Author**

**Date: June 2026**

**Dr. K. Shanti Swarup**
Research advisor
Professor
Department of Electrical Engineering
IIT Madras

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# ABSTRACT

**KEYWORDS** Inverse Estimation, Adaptive Overcurrent Protection, Synchronous Generator, Transformer Differential, IBR Line Differential, Levenberg–Marquardt, Confidence Scoring, Digital Twin, Physics-Informed Neural Network, Wide-Area Protection Coordination, IEC 61850, Adversarial Robustness.

The global power grid is undergoing a fundamental structural transformation: synchronous machines—whose high-inertia fault signatures underpinned a century of deterministic relay design—are being displaced at accelerating pace by Inverter-Based Resources (IBRs). Grid-Following (GFL) and Grid-Forming (GFM) converters are software-controlled current sources whose fault currents are limited to 1.1–2.0 pu by fast sub-cycle current limiters, and whose sequence contributions are determined by firmware rather than physical machine reactance. This paradigm shift systematically invalidates the foundational assumptions of legacy overcurrent, distance, and differential relay algorithms.

This thesis proposes the **Self-Adaptive Model-Based Protection (SAMBP)** architecture for future power systems. The framework treats adaptive protection as an inverse estimation problem: each protection element maintains an explicit physical model of its protected device and identifies model parameters from real-time measurements by the Levenberg–Marquardt algorithm. The framework comprises three tiers and 25 original contributions (17 validated, 8 pending an 18–24 month experimental execution phase).

Chapter 4 develops five adaptive 87L line differential algorithms achieving TPR = 1.000 across $k_{\mathrm{ibr}} \in [0.03, 0.85]$ pu, including 87LN and 87LN0 elements for pure-IBR SLG coverage. Chapter 5 introduces SAMBP-SyncOC: the first inverse-estimation-based adaptive overcurrent protection for synchronous generators, achieving $R^2 > 0.999$ and 60% reduction in spurious relay pickups. Chapter 6 develops the 87T-SPRT transformer

differential scheme using Wald sequential probability ratio test inrush discrimination with explicit false-trip and miss-rate guarantees. Chapters 7–8 validate quadrilateral distance protection and GFL/GFM mode-switching islanding. Chapter 9 delivers a non-blocking EKF digital twin (RMSE $\leq$ 0.024 pu). Chapter 10 presents a physics-informed fault classifier (96.1% accuracy, zero physics violations). Chapter 11 integrates all components into the Wide-Area Protection Coordinator. Chapter 12 establishes the cybersecurity framework including adversarial robustness against CT-injection attacks.

The SAMBP framework is the first published architecture to integrate model-based relay adaptation, digital-twin state estimation, physics-informed classification, and closed-loop coordination across all protection elements with provable safety guarantees.

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# CHAPTER 1

# TEST

Hello world.